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Some set and number properties underlying geometry and physics

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Abstract

Euclidean volume and some distance equations are proved to be derived from ordered combinations (n-tuples) of elements from a list of countable sets. Sequencing a set of n number of elements assigned an immutable linear order in all n! permutations is proved to be cyclic order limited to at most 3 elements: for example, a set of 3 distance dimensions, [width, height, depth], and a set of 3 non-distance dimensions, [time, mass, charge]. Dividing Euclidean distance, time, mass, and charge into n number of constant-sized intervals, for i in 1 to n, there are 3 direct and 3 inverse constant proportion ratios of the i-th constant Euclidean distance interval length to the i-th constant interval lengths of time, mass, and charge. Short, didactic derivations of well-known gravity, charge, electromagnetic, relativity, quantum physics equations and related constants from the ratios and differential calculus are presented. The derivations eliminate the need for dimensional analysis. The inverse proportion ratios are used to add quantum extensions to the Schwarzschild metric, Newton's gravity force, and Coulomb's charge force, which predict that: 1) The gravity and charge constants do not exist where the quantum effects are measurable; 2) The forces between two point masses and between two point charges peak at the Planck unit length.

CONTENTS

Keywords: mathematical physics, gravity, charge, electromagnetism, relativity, quantum.

POPULAR SUMMARY

Euclidean volume and some distance equations are derived from a few set and number properties. Two consequences of the properties of volume and distance are a set of 3 direct and a set of 3 inverse proportion ratios. Using those two sets of ratios (and a little differential calculus), short and didactic derivations of well-known gravity, charge, electromagnetic, relativity, quantum physics equations and related constants are presented.

INTRODUCTION

Insufficient reduction

The philosophy of reductionism posits that complex phenomena can be reduced to simpler components and relations. But, the Einstein field equations (general theory of relativity) and Schrödenger's wave equation (quantum physics) are examples of complicated equations with difficulty in finding exact (analytic) solutions [1–3].

And physics theories often make assumptions without mathematical motivation. For example classical, relativity, and quantum theories assume 3 dimensions of space. But integration, Hilbert spaces, linear algebra, (pseudo-)Riemann manifolds, and so on, have no restriction on the number of dimensions [2, 4–7].

Further, many physics equations rely on volume or distance. But mathematical analysis defines Euclidean volume as the product of interval lengths $\subseteq \mathbb{R}^n$ and defines the Euclidean distance equation, inner product, and metric space [4, 5]. Justifications for the definitions use rigorous finger-pointing to Euclidean geometry, Cartesian geometry, and "It's an abstraction."

The first reduction step, in this article, proves that well-known volume and distance equations are reduced to (derived from) a few simple set and number properties. The second step identifies three properties of the sets and corresponding volume and distance equations that are salient to physics. The third reduction step shows that many well-known physics equations, constants, and assumptions are reduced to (derived from) two consequences of those three properties.

First reduction step

If the underlying principles are simple, then the math proofs should also be simple, trivial even. But even apparently trivial proofs can have subtle flaws. Therefore, each math proof, in this article, has a corresponding formal proof, which has been verified using using the Rocq proof verification system (formerly known as Coq) [8], a verification tool used by mathematicians world-wide. See the data availability section for more details.

Let $|x_i|$ be the cardinal of (number of elements in) the abstract, countable set, $x_i \in \{x_1, \cdots, x_n\}$. And let the value, v_c, be the number of ordered combinations (n-tuples) of the elements of $x_1, \cdots, x_n$. Notionally:

$$\forall\, x_i \in \{x_1, \cdots, x_n\}, \quad \bigcap_{i=1}^{n} x_i = \emptyset, \quad |x_i| \in \{0, \mathbb{N}\}, \quad v_c = \prod_{i=1}^{n} |x_i|. \tag{1}$$

The Euclidean volume equation will be proved to be the trivial case where each n-tuple corresponds to a scalar value, $\kappa \in \mathbb{R}$ and v_c is the total number of n-tuples:

$$v_c \kappa = (\prod_{i=1}^{n} |x_i|)\kappa \quad \Rightarrow \quad \exists\, v, s_i \in \mathbb{R}: \;\; v = \prod_{i=1}^{n} s_i. \tag{2}$$

Note that the goal, here, is to derive the Euclidean volume equation and is *not* a goal to prove that each s_i is an interval length and *not* a goal to prove that $\prod_{i=1}^{n} s_i \subseteq \mathbb{R}^n$.

For all $n > 1$, there are cases where different domain values, $|x_1|, \cdots, |x_n|$, multiplied yield the same range value, v_c. Inferring a domain value, d_c, from the range value, v_c, requires an inverse (bijective) function, $d_c = f_n^{-1}(v_c)$ and $v_c = f_n(d_c)$. The trivial bijective case for all n is the cuboid case, $|x_i| = d_c$. And, where v_c is the sum of the cardinals of subset n-tuples, it will be proved that:

$$d_c^n = v_c = \sum_{i=1}^{m} v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^{m} (\prod_{j=1}^{n} s_{i,j}). \tag{3}$$

The $n = 2$ case is the inner product, where d is the norm [5]:

$$d^2 = \sum_{i=1}^{m} (\prod_{j=1}^{n=2} s_{i,j}) \;\wedge\; \exists\, a_i, b_i \in \mathbb{R}: \;\; s_{i,1} = a_i, \;\; s_{i,2} = b_i \;\Rightarrow\; d^2 = \sum_{i=1}^{m} a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \tag{4}$$

And, where each summed volume, v_{c_i}, is also the bijective function, $d_{c_i}^n = v_{c_i}$:

$$d_c^n = v_c = \sum_{i=1}^{m} v_{c_i} = \sum_{i=1}^{m} d_{c_i}^n \quad \Rightarrow \quad d^n = \sum_{i=1}^{m} d_i^n. \tag{5}$$

$|d|$ is the p-norm (Minkowski distance) [9], which will be proved to imply the metric space properties [4, 5]. The $n = 2$ case is the Euclidean distance (flat distance metric) [10, 11].

Multiplying each summed infinitesimal area, $\mathrm{d}s_i^2$, by a scalar function, $\alpha_i : d^2 = \sum_{i=1}^{m}(\alpha_i \mathrm{d}s_i^2)$, is a Gaussian-like curved distance metric that motivated the metric, $g_{\mu,\nu}$, in Einstein's general relativity theory [1].

Second reduction step

One property salient to physics: All distances derived here are bijective functions of a volume, $v \in \mathbb{R}$, which implies the dimension of each type of distance (for example, Euclidean distance) is always a $\subseteq \mathbb{R}$. *Note* that distance is *not* a bijective function of the volume domain values, for example: width, height, and depth.

A second property salient to physics: The commutative property of the multiplication operations defining the number n-tuples and corresponding Euclidean volume (2) allow n number of n-tuple domain set elements and volume domain values to be sequenced in all $n!$ permutations, for example, $v = \mathit{width} \times \mathit{height} \times \mathit{depth} = \mathit{depth} \times \mathit{width} \times \mathit{height} = \cdots$. That is, the n-tuple domain set elements and volume domain values are unordered (no element has an intrinsic position property).

A third property salient to physics: All the set elements, $x_1, \cdots, x_n$ and volume domain values, $s_1, \cdots, s_n$, must have the property of being linearly sequenced via successor and predecessor functions, for example, $\mathit{successor}(x_i) = x_{i+1}$ and $\mathit{predecessor}(x_i) = x_{i-1}$.

Repeatable sequencing of unordered elements requires assigning an immutable linear order (an order that does not change). Where the order of the set elements is immutable, only a cyclic order (list), where $\mathit{successor}(x_n) = x_1$ and $\mathit{predecessor}(x_1) = x_n$, allows starting at each element and linearly sequencing n elements in a successor permutation and a predecessor permutation. For example, where $n = 3$, the volume cyclic successor and predecessor sequences are: $[s_1, s_2, s_3]$ and $[s_1, s_3, s_2]$; $[s_2, s_3, s_1]$ and $[s_2, s_1, s_3]$; $[s_3, s_1, s_2]$ and $[s_3, s_2, s_1]$.

The only cyclic lists that allows the linear sequencing of n number of elements in all $n!$ permutations, is where each list element is cyclic adjacent (either an *immediate* cyclic successor or an *immediate* cyclic predecessor) to every other element, referred to, in this article, as an All Adjacent Cyclic List (AACL). For example, the cyclic permutations, $[x_2, x_3, x_1]$ and $[x_2, x_1, x_3]$, require x_1 and x_3 be cyclic adjacent. An AACL will be proved to contain $n \leq 3$ elements.

Third reduction step

Application of the Euclidean distance dimension $\subseteq \mathbb{R}$ property, the commutative and linearly sequenced properties of volume domain elements to physics: There is an AACL of 3 distance dimensions and an AACL of 3 non-distance dimensions (time, mass, and charge), each dimension $\subseteq \mathbb{R}$.

Where the dimensions Euclidean distance, time, mass and charge are divided into the same number of constant-sized intervals, each constant Euclidean distance interval length corresponds to constant interval lengths of time, mass, and charge. As a consequence there are a set of 3 constant direct proportion ratios and a set of 3 constant inverse proportion ratios.

The gravity [12, 13], special relativity [1, 2], Schwarzschild black hole metric [14, 15], charge equations [13], electromagnetism equations [13], and related constants are derived from the 3 direct proportion ratios. The Planck-Einstein relation [13], Compton wavelength [13], Schrödenger wave equation [3], Dirac wave equation [16], and related constants are derived from combining some of the direct and inverse proportion ratios. And the inverse proportion ratios are used to add quantum extensions to Scharwzchild's metric, Newton's gravity force equation, and Coulomb's charge force equation.

VOLUME

Euclidean volume

Theorem .1. *Euclidean volume,*

$$\forall\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\},\ v_c = \textstyle\prod_{i=1}^n |x_i| \ \Rightarrow\ v = \prod_{i=1}^n s_i, \quad s_i, v \in \mathbb{R}. \quad (6)$$

The formal proof, "Euclidean_volume," is in the Rocq file, euclidrelations.v.

Proof.

$$\forall v_c, |x_i| \in \{0, \mathbb{N}\},\ \ \kappa \in \mathbb{R}:\ \ v_c = \textstyle\prod_{i=1}^n |x_i| \ \Leftrightarrow\ v_c\kappa = (\prod_{i=1}^n |x_i|)\kappa = \prod_{i=1}^n (|x_i|\kappa^{1/n}). \quad (7)$$

$$v_c\kappa = \textstyle\prod_{i=1}^n (|x_i|\kappa^{1/n}) \ \wedge\ \exists\, v, s_i \in \mathbb{R}:\ \ v = v_c\kappa \ \wedge\ s_i = |x_i|\kappa^{1/n} \ \Rightarrow\ v = \prod_{i=1}^n s_i. \quad \square \quad (8)$$

Lemma .2. *The number of n-tuples, v_c, is the sum of the number of n-tuples, v_{c_i}, in each disjoint subset of n-tuples, implies a volume is the sum of volumes,*

$$v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad v = \sum_{i=1}^m v_i, \quad v, v_i \in \mathbb{R}.$$

The formal proof, "sum_of _volumes," is in the Rocq file, euclidrelations.v.

Proof. From the condition of this theorem and the Euclidean volume theorem (.1):

$$\forall\, \kappa \in \mathbb{R}: \ v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad v_c\kappa = (\sum_{i=1}^m v_{c_i})\kappa = \sum_{i=1}^m (v_{c_i}\kappa). \tag{9}$$

$$v_c\kappa = \sum_{i=1}^m (v_{c_i}\kappa) \quad \wedge \quad v = v_c\kappa \quad \wedge \quad v_i = v_{c_i}\kappa \quad \Rightarrow \quad v = \sum_{i=1}^m v_i. \quad \square \tag{10}$$

DISTANCE

Definition .3. Bijective, countable domain value, d_c:

$$\exists\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\}, \quad v_c = \prod_{i=1}^n |x_i| = \prod_{i=1}^n d_c = d_c^n. \tag{11}$$

Inner product

Theorem .4. *Sum of volumes distance:*

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i_j}).$$

The formal proof, "sum_of _volumes _distance," is in the Rocq file, euclidrelations.v.

Proof. Multiply both sides of the assumption by κ and apply the Euclidean volume proof (.1) to v_i:

$$v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad v = \sum_{i=1}^m v_i, \quad v, v_i \in \mathbb{R}. \tag{12}$$

$$v = \sum_{i=1}^m v_i \quad \wedge \quad v_i = \prod_j^n s_{i,j} \quad \Rightarrow \quad v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \tag{13}$$

$$d_c^n = v_c \quad \Rightarrow \quad d_c^n\kappa = (d_c\kappa^{1/n})^n = v_c\kappa. \tag{14}$$

$$\exists\, d, v \in \mathbb{R} : d = d_c\kappa^{1/n} \quad \wedge \quad v = v_c\kappa \quad \wedge \quad (d_c\kappa^{1/n})^n = v_c\kappa \quad \Rightarrow \quad d^n = v. \tag{15}$$

$$d^n = v \quad \wedge \quad v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}) \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \quad \square \tag{16}$$

The $n = 2$ case is the inner product, where d is the norm [5]:

$$d^2 = \sum_{i=1}^m (\prod_{j=1}^{n=2} s_{i,j}) \wedge \exists\, a_{i,1}, a_{i,2} \in \mathbb{R}: \ s_{i,1} = a_i,\ s_{i,2} = b_i \ \Rightarrow\ d^2 = \sum_{i=1}^m a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \tag{17}$$

The inner product also allows a_i and b_i to be complex numbers, functions, and matrices.

Minkowski distance (p-norm)

Theorem .5. *Minkowski distance (p-norm):*

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} = \sum_{i=1}^m d_{c_i}^n \quad \Leftrightarrow \quad d^n = \sum_{i=1}^m d_i^n.$$

The formal proof, "Minkowski_distance," is in the Rocq file, euclidrelations.v.

Proof. From the sum of volumes distance theorem .4:

$$d_{c_i}^n = v_{c_i} \quad \Rightarrow \quad d_i^n = v_i. \tag{18}$$

$$d^n = v = \sum_{i=1}^m v_i \quad \wedge \quad d_i^n = v_i \Rightarrow \quad d^n = \sum_{i=1}^m d_i^n \quad \square \tag{19}$$

Distance inequalities

Theorem .6. *Distance triangle inequality*

$$\forall\, n \in \mathbb{N},\ \ v_a, v_b \geq 0: \ \ (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}.$$

The formal proof, distance_inequality, is in the Rocq file, euclidrelations.v.

Proof. Expand $(v_a^{1/n} + v_b^{1/n})^n$ using the binomial expansion:

$$\begin{aligned}\forall\, v_a, v_b \geq 0: \quad v_a + v_b \leq v_a + v_b + \textstyle\sum_{i=1}^{n} \binom{n}{k} (v_a^{1/n})^{n-k} (v_b^{1/n})^k \\ + \textstyle\sum_{i=1}^{n} \binom{n}{k} (v_a^{1/n})^k (v_b^{1/n})^{n-k} = (v_a^{1/n} + v_b^{1/n})^n.\end{aligned} \tag{20}$$

Take the n^{th} root of both sides of the inequality 20:

$$\forall\, v_a, v_b \geq 0,\ \ n \in \mathbb{N}: \quad v_a + v_b \leq (v_a^{1/n} + v_b^{1/n})^n \quad \Rightarrow \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}. \tag{21}$$

$\square$

Theorem .7. *Distance sum inequality*

$$\forall\, m, n \in \mathbb{N},\ \ a_i, b_i \geq 0: (\textstyle\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}.$$

The formal proof, distance_sum_inequality, is in the Rocq file, euclidrelations.v.

Proof. Apply the distance triangle inequality (.6):

$$\begin{aligned}&\forall\, m, n \in \mathbb{N},\ \ v_a, v_b \geq 0: \ v_a = \textstyle\sum_{i=1}^m a_i^n \quad \wedge \quad v_b = \sum_{i=1}^m b_i^n \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n} \\ &\Rightarrow \quad ((\textstyle\sum_{i=1}^m a_i^n) + (\sum_{i=1}^m b_i^n))^{1/n} = (\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}. \quad \square\end{aligned} \tag{22}$$

Metric Space

All Minkowski distances (p-norms) imply the metric space properties. The formal proofs: triangle_inequality, symmetry, non_negativity, and identity_of_indiscernibles are in the Rocq file, euclidrelations.v.

Theorem .8. *Triangle Inequality:*

$$d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$$
$$\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w).$$

Proof. $\forall\, p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$:

$$(u^p + w^p)^{1/p} \leq ((u^p + w^p) + 2v^p)^{1/p} = ((u^p + v^p) + (v^p + w^p))^{1/p}. \tag{23}$$

Apply the distance triangle inequality (.6) to the inequality 23:

$$(u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}$$
$$\wedge \quad v_a = u^p + v^p \quad \wedge \quad v_b = v^p + w^p$$
$$\Rightarrow \quad (u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p}$$
$$\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w). \quad \square \tag{24}$$

Theorem .9. *Symmetry:*
$d(s_1, s_2) = (\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad v = s_2 \quad \Rightarrow \quad d(u, v) = d(v, u).$

Proof. By the commutative law of addition:

$$\forall\, p \,:\, p \geq 1, \quad d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p} = (s_1^p + s_2^p)^{1/p}$$
$$\Rightarrow \quad d(u, v) = (u^p + v^p)^{1/p} = (v^p + u^p)^{1/p} = d(v, u). \quad \square \tag{25}$$

Theorem .10. *Non-negativity:*

$$d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \in N \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(u, v) \geq 0.$$

Proof.

$$\forall\, n \geq 1 \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(s_1, s_2) = (s_1^p + s_1^p)^{1/p} \geq 0. \tag{26}$$

$$\forall\, u = s_1,\ v = s_2 \ \wedge\ d(s_1, s_2) = (s_1^p + s_1^p)^{1/p} \geq 0 \ \Rightarrow\ d(u, v) = (u^p + v^p)^{1/p} \geq 0. \quad \square \tag{27}$$

Theorem .11. *Identity of Indiscernibles:* $d(u,u) = 0$.

Proof. From the non-negativity property (.10):

$$d(u,w) \geq 0 \quad \wedge \quad d(u,v) \geq 0 \quad \wedge \quad d(v,w) \geq 0 \tag{28}$$

$$\Rightarrow \quad \exists\, d(u,w) = d(u,v) = d(v,w) = 0. \tag{29}$$

$$d(u,w) = d(v,w) = 0 \quad \Rightarrow \quad u = v. \tag{30}$$

$$d(u,v) = 0 \quad \wedge \quad u = v \quad \Rightarrow \quad d(u,u) = 0. \quad \square \tag{31}$$

PROPERTIES LIMITING A SET TO AT MOST 3 ELEMENTS

The following definitions and proof use first order logic. A Horn clause-like expression is used, here, to make the proof easier to read. By convention, the proof goal, in a Horn clause, is on the left side of the implication sign ($\leftarrow$) and supporting facts are on the right side. The formal proofs in the Rocq file, threed.v, are: adj111, adj122, adj212, adj123, adj133, adj233, adj213, adj313, adj323, and not_all_mutually_adjacent_gt_3.

Definition .12. Immediate Cyclic Successor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$

$$Successor(m,n,setsize) \;\leftarrow\; (m = setsize \wedge n = 1) \;\vee\; (n = m + 1 \wedge n \leq setsize). \tag{32}$$

Definition .13. Immediate Cyclic Predecessor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$

$$Predecessor(m,n,setsize) \;\leftarrow\; (m = 1 \wedge n = setsize) \;\vee\; (n = m - 1 \wedge n \geq 1). \tag{33}$$

Definition .14. Adjacent: Element m is sequentially cyclic adjacent to element n if the immediate cyclic successor of m is n or the immediate cyclic predecessor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$

$$Adjacent(m,n,setsize) \leftarrow Successor(m,n,setsize) \;\vee\; Predecessor(m,n,setsize). \tag{34}$$

Definition .15. All Adjacent Cyclic List (AACL), where every set element is cyclic adjacent to every other element):

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} : \quad Adjacent(m,n,setsize). \tag{35}$$

Theorem .16. *An AACL is limited to at most 3 elements.*

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} : \quad Adjacent(m, n, setsize) \quad \Rightarrow \quad setsize \leq 3. \tag{36}$$

Proof.

Every element is adjacent to every other element, where $setsize \in \{1,\ 2,\ 3\}$:

$$Adjacent(1,1,1) \leftarrow Successor(1,1,1) \leftarrow (m = setsize \wedge n = 1). \tag{37}$$

$$Adjacent(1,2,2) \leftarrow Successor(1,2,2) \leftarrow (n = m + 1 \wedge n \leq setsize). \tag{38}$$

$$Adjacent(1,2,3) \leftarrow Successor(1,2,3) \leftarrow (n = m + 1 \wedge n \leq setsize). \tag{39}$$

$$Adjacent(2,1,3) \leftarrow Predecessor(2,1,3) \leftarrow (n = m - 1 \wedge n \geq 1). \tag{40}$$

$$Adjacent(3,1,3) \leftarrow Successor(3,1,3) \leftarrow (n = setsize \wedge m = 1). \tag{41}$$

$$Adjacent(1,3,3) \leftarrow Predecessor(1,3,3) \leftarrow (m = 1 \wedge n = setsize). \tag{42}$$

$$Adjacent(2,3,3) \leftarrow Successor(2,3,3) \leftarrow (n = m + 1 \wedge n \leq setsize). \tag{43}$$

$$Adjacent(3,2,3) \leftarrow Predecessor(3,2,3) \leftarrow (n = m - 1 \wedge n \geq 1). \tag{44}$$

Element 2 is the only immediate successor of element 1 for all $setsize \geq 3$, which implies element 3 is not ($\neg$) an immediate successor of element 1 for all $setsize \geq 3$:

$$\neg Successor(1,3,setsize \geq 3) \\ \leftarrow Successor(1,2,setsize \geq 3) \leftarrow (n = m + 1 \leq setsize). \tag{45}$$

Element $n = setsize > 3$ is the only immediate predecessor of element 1, which implies element 3 is not ($\neg$) an immediate predecessor of element 1 for all $setsize > 3$:

$$\neg Predecessor(1,3,setsize \geq 3) \\ \leftarrow Predecessor(1,setsize,setsize > 3) \leftarrow (m = 1 \wedge n = setsize > 3). \tag{46}$$

For all $setsize > 3$, some elements are not ($\neg$) sequentially adjacent to every other element (not immediate symmetric):

$$\neg Adjacent(1,3,setsize > 3) \\ \leftarrow \neg Successor(1,3,setsize > 3) \quad \wedge \quad \neg Predecessor(1,3,setsize > 3). \quad \square \tag{47}$$

The Symmetric goal matches the Adjacent goals 37 and fails for all "setsize" greater than three.

DERIVATION OF PHYSICS EQUATIONS AND CONSTANTS

3 direct and 3 inverse proportion ratios

Application of the AACL property of volume (.16) and the Euclidean distance dimension $\subseteq \mathbb{R}$ to physics (.5): $\{r_1, r_2, r_3\}$ is an AACL of 3 "distance" dimensions, each dimension having the $\subseteq \mathbb{R}$ set membership property and $\{t\ (time), m\ (mass),\ q\ (charge)\}$ is an AACL of 3 "non-distance" dimensions, each dimension $\subseteq \mathbb{R}$. There is a total of 6 dimensions with the $\subseteq \mathbb{R}$ property: $r_1,\ r_2,\ r_3,\ t,\ m,\ q$.

Where the dimensions of Euclidean distance, time, mass, and charge are each divided into n number of constant-sized interval lengths, for i in 1 to n, the i^{th} constant unit interval length, r_p, of Euclidean distance corresponds to the i^{th} constant unit interval lengths: t_p of time, m_p of mass, and q_p of charge, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

Two consequences of the AACL and corresponding constant interval lengths:

1. A set of 3 constant direct proportion ratios and proportion constants:

$$\forall\ r_p, t_p, m_p, q_p, k_1, k_2, k_3 \in \mathbb{R},\ \exists\ r, t, m, q \in \mathbb{R}:$$
$$\frac{r_p}{t_p} = \frac{r_p k_1}{t_p k_1} = \frac{r}{t} \quad \wedge \quad \frac{r_p}{m_p} = \frac{r_p k_2}{m_p k_2} = \frac{r}{m} \quad \wedge \quad \frac{r_p}{q_p} = \frac{r_p k_3}{q_p k_3} = \frac{r}{q}. \tag{48}$$

$$\exists\ c, c_t, c_m, c_q \in \mathbb{R}: \quad r_p/t_p = c_t = c \quad \wedge \quad r_p/m_p = c_m \quad \wedge \quad r_p/q_p = c_q$$
$$\Rightarrow \quad r/t = r_p/t_p = c_t = c \quad \wedge \quad r/m = r_p/m_p = c_m \quad \wedge \quad r/q = r_p/q_p. = c_q. \tag{49}$$

2. A set of 3 constant inverse proportion ratios and proportion constants:

$$\forall\ r_p, t_p, m_p, q_p, k_1, k_2, k_3 \in \mathbb{R},\ \exists\ r, t, m, q \in \mathbb{R}:$$
$$r_p t_p = r_p t_p \left(\frac{k_1}{k_1}\right) = \left(\frac{r_p}{k_1}\right)(t_p k_1) = rt \quad \wedge \quad r_p m_p = r_p m_p \left(\frac{k_2}{k_2}\right) = \left(\frac{r_p}{k_2}\right)(m_p k_2) = rm$$
$$\wedge \quad r_p q_p = r_p q_p \left(\frac{k_3}{k_3}\right) = \left(\frac{r_p}{k_3}\right)(q_p k_3) = rq. \tag{50}$$

$$\exists\ k_t, k_m, k_q \in \mathbb{R}: \quad r_p t_p = k_t \quad \wedge \quad r_p m_p = k_m \quad \wedge \quad r_p q_p = k_q$$
$$\Rightarrow \quad rt = r_p t_p = k_t \quad \wedge \quad rm = r_p m_p = k_m \quad \wedge \quad rq = r_p q_p = k_q. \tag{51}$$

G, Newton, Gauss, Poisson gravity laws

From equations 49, linear acceleration, r/t^2, can be related to mass, m, and distance, r:

$$r = c_m m \ \wedge \ r = c_t t \ \Rightarrow \ \frac{r}{(c_t t)^2} = \frac{c_m m}{r^2} \ \Rightarrow \ \frac{r}{t^2} = \frac{(c_m c_t^2)m}{r^2} = \frac{Gm}{r^2}. \tag{52}$$

Where r_p, t_p, and m_p are the Planck length, time, and mass units, $G = c_m c_t^2 = r_p^3/m_p t_p^2$ $\approx 6.67430 \cdot 10^{-11}\ m^3 \cdot kg^{-1} \cdot s^{-2}$, which are the empirical value and SI units of G [17].

Newton's gravity law [12] follows from multiplying both sides of equation 52 by m:

$$\frac{r}{t^2} = \frac{Gm}{r^2} \ \Leftrightarrow \ F \stackrel{\text{def}}{=} \frac{mr}{t^2} = \frac{Gm^2}{r^2}. \tag{53}$$

$$F = \frac{Gm^2}{r^2} \ \wedge \ \forall\, m \in \mathbb{R}:\ \exists\, m_1, m_2 \in \mathbb{R}:\ m_1 m_2 = m^2 \quad \Rightarrow \quad F = \frac{Gm_1 m_2}{r^2}. \tag{54}$$

A new interpretation of Gauss's and Poisson's gravity laws is presented here. Equation 52 relates linear acceleration, r/t^2, to mass and distance. Gauss's gravity vector field, $\mathbf{g}$, and Poisson's gravity scalar field, $-\nabla\Phi$, is the angular acceleration, $2\pi r/t^2$. Applying equation 52, where the minus sign is the vector direction (attractive force):

$$\frac{r}{t^2} = -\frac{Gm}{|\mathbf{r}|^2} \ \wedge \ \mathbf{g} = -\nabla\Phi = \frac{2\pi r}{t^2} \ \Rightarrow \ \mathbf{g} = -\nabla\Phi = -\frac{2\pi Gm}{|\mathbf{r}|^2}. \tag{55}$$

$$\mathbf{g} = -\nabla\Phi = -\frac{2\pi Gm}{|\mathbf{r}|^2} \quad \Rightarrow \quad \nabla\cdot\mathbf{g} = \nabla^2\Phi = \frac{4\pi Gm}{|\mathbf{r}|^3} = 4\pi G\rho, \tag{56}$$

which are the differential forms of the Gauss and Poisson gravity laws [13], where $\rho = m/|\mathbf{r}|^3$ is the mass density:

Special relativity

$$\forall\, \tau \in \{t, m, q\},\, r^2 = r'^2 + r_v^2,\, \exists\, \mu_\tau, \nu_\tau : r = \mu_\tau\tau \quad \wedge \quad r_v = \nu_\tau\tau \quad \Rightarrow \quad (\mu_\tau\tau)^2 = r'^2 + (\nu_\tau\tau)^2$$
$$\Rightarrow \quad r' = \sqrt{(\mu\tau)^2 - (\nu_\tau\tau)^2} = \mu_\tau\tau\sqrt{1 - (\nu_\tau/\mu_\tau)^2}. \tag{57}$$

Local frame distance, r', contracts relative to a distant observer frame distance, r, as $\nu_\tau \to \mu_\tau$:

$$r' = \mu_\tau\tau\sqrt{1 - (\nu_\tau/\mu_\tau)^2} \quad \wedge \quad \mu_\tau\tau = r \quad \Rightarrow \quad r' = r\sqrt{1 - (\nu_\tau/\mu_\tau)^2}. \tag{58}$$

Where τ' is time in the local frame, less time lapses (time dilates) relative to time, τ, in the distant (coordinate) frame: τ, as $\nu_\tau \to \mu_\tau$:

$$\mu_\tau \tau = r'/\sqrt{1-(\nu_\tau/\mu_\tau)^2} \quad \wedge \quad r' = \mu_\tau \tau \quad \Rightarrow \quad \tau' = \tau\sqrt{1-(\nu_\tau/\mu_\tau)^2}. \tag{59}$$

Where τ is type, time, the space-like flat Minkowski spacetime event interval is:

$$\mathrm{d}r^2 = \mathrm{d}r'^2 + \mathrm{d}r_v^2 \quad \wedge \quad \mathrm{d}r_v^2 = \mathrm{d}r_1^2 + \mathrm{d}r_2^2 + \mathrm{d}r_3^2 \quad \wedge \quad \mathrm{d}(\mu_\tau\tau) = \mathrm{d}r$$
$$\Rightarrow \ \mathrm{d}r'^2 = \mathrm{d}(\mu_\tau\tau)^2 - \mathrm{d}r_1^2 - \mathrm{d}r_2^2 - \mathrm{d}r_3^2. \tag{60}$$

Analytic general relativity solutions

The Einstein field equation (EFE) is:

$$\mathbf{G}_{\mu,\nu} + \Lambda g_{\mu,\nu} = \kappa T_{\mu,\nu}, \quad \mathbf{G}_{\mu,\nu} = \mathbf{R} + \frac{1}{2} R g_{\mu,\nu}. \tag{61}$$

where: $\mathbf{G}_{\mu,\nu}$ is Einstein's tensor, $\mathbf{R}$ is the Ricci curvature, R is the scalar curvature, $g_{\mu,\nu}$ is the metric tensor, Λ is a cosmological constant, $\kappa = 8\pi G/c^4$, is Einstein's constant, and $T_{\mu,\nu}$ is the stress-energy-momentum tensor [1, 2].

The EFE is a set of second order differential equations extending the previously derived second order differential equation, Poisson's law of gravity, $\nabla^2\Phi = 4\pi G\rho$ (56), to include a time component. *Note*: 1) The Minkowski spacetime interval component for time is $c^2 g_{0,0}\mathrm{d}t^2$ [2]. 2) By convention, $T_{0,0} = mc^2/|\mathbf{r}|^3$ is the energy density component of $T_{\mu,\nu}$ [2].

$$c^2 g_{0,0} \approx \frac{c^2}{2}(\mathbf{G}_{0,0} + \Lambda g_{0,0}) = \nabla^2\Phi = 4\pi G\rho = 4\pi G\frac{m}{|\mathbf{r}|^3} = \frac{4\pi G}{c^2}\cdot\frac{mc^2}{|\mathbf{r}|^3} = \frac{4\pi G}{c^2}T_{0,0}. \tag{62}$$
$$\frac{c^2}{2}(\mathbf{G}_{0,0} + \Lambda g_{0,0}) = \frac{4\pi G}{c^2}T_{0,0} \quad \Rightarrow \quad \mathbf{G}_{0,0} + \Lambda g_{0,0} = \frac{8\pi G}{c^4}T_{0,0} = \kappa T_{0,0}. \tag{63}$$

1) The laws of physics determine the amount of spacetime curvature and the distribution of stress, energy, and momentum. 2) The laws of physics are specified as the components of the metric, $g_{\mu,\nu}$. 3) $g_{\mu,\nu}$ can be determined independent of the EFE, for those cases where the laws (equations) can be derived from the ratios. Derivation of the Schwarzschild metric from the ratios, in the next subsection, is an example.

Step 1) Use the unit ratios and equations (laws) derived from the ratios, for example, the ratio-derived special relativity equations (58) and the ratio-derived constant, G (52), to derive the scalar functions (laws) assigned to the components of $g_{\mu,\nu}$.

Step 2) The 4D flat spacetime interval equation is an instance of the 2D equation (60). Therefore, derive the metric first as a 2D metric and expand later to a 4D metric.

(Optional step 2d) The 2D metric tensor uses the much simpler 2D Ricci curvature (which, in 2D, is the Gaussian curvature). And the 2D tensors reduce the number of independent equations to solve. The 2D solutions can next be used to set constraints on the solutions in the 4D tensor, much like an Euclidean distance value places constraints on the coordinate distance values.

Step 3) Translate from 2D to 4D. For spherically symmetric cases, the simplest method to translate from 2D to 4D is to use spherical coordinates, where the first 2 diagonal components of $g_{\nu,\mu}$ remain unchanged and the other 2 diagonal components are functions of the longitude angle, θ, and latitude angle, ϕ.

Schwarzschild's time dilation and metric

From equations 58 and 49:

$$r = c_m m \quad \Rightarrow \quad c_m m/r = 1 \quad \Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-(1)(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2}. \tag{64}$$

Where v_{escape} is the escape velocity and KE is the kinetic energy:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv^2_{escape} \\ \Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v^2_{escape}/rc^2}. \tag{65}$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v^2_{escape}/rc^2}. \quad \wedge \quad v_{escape} = c \\ \Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m mc^2/rc^2}. \tag{66}$$

Combining equation 66 with the derivation of G (52):

$$c_m c^2 = G \wedge \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m mc^2/rc^2} \Rightarrow \sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2}. \tag{67}$$

Schwarzschild defined the black hole event horizon radius as $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$ [14, 15]. Today, α, is called the Schwarzschild radius, $r_s \stackrel{\text{def}}{=} 2Gm/c^2$:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2} \quad \wedge \quad r_s \stackrel{\text{def}}{=} 2Gm/c^2 \quad \Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-r_s/r}. \tag{68}$$

Combining equation 68 with equation 59 yields Schwarzschild's gravitational time dilation:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-r_s/r} \quad \wedge \quad t' = t\sqrt{1-(v^2/c^2)} \quad \Rightarrow \quad t' = t\sqrt{1-r_s/r}. \tag{69}$$

From equations 58 and 68.

$$r' = r\sqrt{1-(v/c)^2} \quad \wedge \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-r_s/r} \quad \Rightarrow \quad r' = r\sqrt{1-r_s/r}. \tag{70}$$

Applying equation 70 to the time-like spacetime interval equation 60, where $\mathrm{d}s^2 < 0$:

$$r' = r\sqrt{1-r_s/r} \quad \wedge \quad \mathrm{d}s^2 = \mathrm{d}r'^2 - \mathrm{d}r^2$$

$$\Rightarrow \quad \mathrm{d}s^2 = (\sqrt{1-r_s/r}\mathrm{d}r)^2 - (\mathrm{d}r'/\sqrt{1-r_s/r})^2 = (1-r_s/r)\mathrm{d}r^2 - (1-r_s/r)^{-1}\mathrm{d}r'^2. \tag{71}$$

$$\mathrm{d}s^2 = (1-r_s/r)\mathrm{d}r^2 - (1-r_s/r)^{-1}\mathrm{d}r'^2 \quad \wedge \quad \mathrm{d}r = \mathrm{d}(ct) \quad \wedge \quad c = 1$$

$$\Rightarrow \quad \mathrm{d}s^2 = (1-r_s/r)\mathrm{d}t^2 - (1-r_s/r)^{-1}\mathrm{d}r'^2. \tag{72}$$

Using spherical coordinates to translate from 2D to 4D yields the $+---$ form of Schwarzschild's black hole metric, $g_{\mu,\nu}$ [14, 15]:

$$(1-r_s/r)\mathrm{d}t^2-(1-r_s/r)^{-1}\mathrm{d}r'^2 \Rightarrow \mathrm{d}s^2 = (1-r_s/r)\mathrm{d}t^2-(1-r_s/r)^{-1}\mathrm{d}r'^2-r^2(\mathrm{d}\theta^2+sin^2\theta\mathrm{d}\phi^2)$$

$$\Rightarrow \quad g_{\mu,\nu} = \mathrm{diag}[(1-r_s/r),\ -(1-r_s/r)^{-1},\ -r^2,\ -r^2sin^2\theta]. \tag{73}$$

k_e, ε_0, E, Coulomb and Gauss laws

From equations 49, linear acceleration, r/t^2, can be related to charge, q, and distance, r:

$$r = c_q q \ \wedge \ r = c_t t \ \Rightarrow \ \frac{r}{(c_t t)^2} = \frac{c_q q}{r^2} \ \Rightarrow \ \frac{r}{t^2} = \frac{(c_q c_t^2)q}{r^2}. \tag{74}$$

$$r = c_m m \quad \wedge \quad r = c_q q \quad \Rightarrow \quad c_m m = c_q q \quad \Rightarrow \quad m = (c_q/c_m)q. \tag{75}$$

Combining equations 74 and 75 yields Coulombs's law and constant, k_e,[13]:

$$\frac{r}{t^2} = \frac{(c_q c_t^2)q}{r^2} \ \wedge \ m = (c_q/c_m)q \ \Rightarrow \ F \stackrel{\text{def}}{=} \frac{mr}{t^2} = \frac{(c_{q^2}c_t^2/c_m)q^2}{r^2} = \frac{k_e q^2}{r^2}. \tag{76}$$

Where r_p, t_p,, m_p, and q_p are the Planck units, $k_e = c_q^2 c_t^2/c_m = m_p r_p^3/t_p^2 q_p^2$, has the empirical value and SI units: $kg \cdot m^3 \cdot s^{-2} \cdot C^{-2} \stackrel{\text{def}}{=} N \cdot m^2 \cdot C^{-2}$ [13, 17].

$$\forall\, q \in \mathbb{R}\ \exists\ q_1, q_2 \in \mathbb{R} :\ |q_1 q_2| = q^2 \quad \wedge \quad F = \frac{k_e q^2}{r^2}. \quad \Rightarrow \quad F = \frac{k_e|q_1 q_2|}{r^2}. \tag{77}$$

A new interpretation of Gauss's electric field law is presented here. Equation 74 relates the linear acceleration, r/t^2, to charge and distance. Gauss's electric field, $\mathbf{E}$, relates angular acceleration, $2\pi r/t^2$ to charge and distance. Applying equation 76:

$$F_C = \frac{k_e q^2}{r^2} \quad \Rightarrow \quad \exists\, F_E \in \mathbb{R}:\ F_E = 2\pi F_C = \frac{2\pi k_e q^2}{r^2}. \tag{78}$$

$$F_E = \frac{2\pi k_e q^2}{r^2} = q\left(\frac{2\pi k_e q}{r^2}\right) \quad \wedge \quad \exists\, E \in R:\ E = \frac{2\pi k_e q}{r^2} \quad \Rightarrow \quad F_E = qE. \tag{79}$$

The electric field, $E = 2\pi k_e q/r^2$, conforms to the SI units $kg \cdot m \cdot s^{-2} \cdot C^{-1} \stackrel{\text{def}}{=} kg \cdot m \cdot s^{-3} \cdot A^{-1}$ [17].

$$\mathbf{E} = \frac{2\pi k_e q}{|\mathbf{r}|^2} \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\frac{4\pi k_e q}{|\mathbf{r}|^3}. \tag{80}$$

$$\nabla \cdot \mathbf{E} = -\frac{4\pi k_e q}{|\mathbf{r}|^3} \quad \wedge \quad \varepsilon_0 \stackrel{\text{def}}{=} \frac{1}{4\pi k_e} \quad \wedge \quad \rho \stackrel{\text{def}}{=} \frac{q}{|\mathbf{r}|^3} \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\rho/\varepsilon_0, \tag{81}$$

which is Gauss's electric field law in differential form [13].

B, μ_0, and Lorentz's law

Applying the distance contraction equation, 58, to equation 79, where r' and r are swapped such that r' is the distant observer frame and r is local (rest) frame of reference:

$$r' = r/\sqrt{1 - v^2/c^2} \quad \wedge \quad F' = \frac{2\pi k_e q^2}{r'^2} \quad \Rightarrow \quad F = \frac{2\pi k_e q^2 (1 - v^2/c^2)}{r^2}. \tag{82}$$

From equation 79:

$$E = \frac{2\pi k_e q}{r^2} \ \wedge \ F = \frac{2\pi k_e q^2(1 - v^2/c^2)}{r^2} \ \Rightarrow \ F = q(E - \frac{(2\pi k_e/c^2)q}{r^2} v^2). \tag{83}$$

$$F = q(E - \frac{(2\pi k_e/c^2)q}{r^2} v^2) \ \wedge \ \exists\, B:\ B = \frac{(2\pi k_e/c^2)vq}{r^2} \ \Rightarrow \ F = q(E - Bv), \tag{84}$$

where the magnetic field, $B = (2\pi k_e/c)vq/r^2$, conforms to the base SI units: $kg \cdot C^{-1} \cdot s^{-1} \stackrel{\text{def}}{=} kg \cdot A^{-1} \cdot s^{-2} \stackrel{\text{def}}{=} T$ [17]. And assigning the constant portion to μ_0:

$$B = \frac{(2\pi k_e/c^2)vq}{r^2} \quad \Rightarrow \quad \frac{\partial B}{\partial r} = \frac{(4\pi k_e/c^2)vq}{r^3} = \frac{\mu_0 vq}{r^3}. \tag{85}$$

where $\mu_0 = 4\pi k_e/c^2 = m_p r_p/q_p^2$ conforms to the SI units, $kg \cdot m \cdot C^{-2} \stackrel{\text{def}}{=} kg \cdot m \cdot s^{-2} A^{-2} \stackrel{\text{def}}{=} N \cdot A^{-2}$ [17]. And translating equation 84 to vector form:

$$F = q(E - Bv) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \mathbf{v}). \tag{86}$$

$$\mathbf{B} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{B})) \quad \wedge \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \mathbf{v}) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \tag{87}$$

which is Lorentz law in the local frame of reference [13].

Faraday's law

From the magnetic field equation 87, where the electric and magnetic fields are propagating at the speed, $v = c$:

$$B = \frac{(2\pi k_e/c^2)vq}{r^2} \quad \wedge \quad v = c \quad \wedge \quad r = ct \quad \Rightarrow \quad B = \frac{(2\pi k_e/c^3)q}{t^2}. \tag{88}$$

$$B = \frac{(2\pi k_e/c^3)q}{t^2} \quad \Rightarrow \quad \frac{\partial B}{\partial t} = -\frac{(4\pi k_e/c^3)q}{t^3}. \tag{89}$$

$$\frac{\partial B}{\partial t} = -\frac{(4\pi k_e/c^3)q}{t^3} \quad \wedge \quad r = ct \quad \Rightarrow \quad \frac{\partial B}{\partial t} = -\frac{4\pi k_e q}{r^3}. \tag{90}$$

Faraday's law is the case, where the electric field, $\mathbf{E}$, is a single radial vector, $\mathbf{E_{r_1}}$ acting on some point, P. Therefore, at point, P, $\mathbf{E_{r_2}} = 0$ and $\mathbf{E_{r_3}} = 0$. From the Lorentz equation (87), the magnetic field, $\mathbf{B}$ at point P, is perpendicular to $\mathbf{E_{r_1}}$ (for example, aligned on $\mathbf{r}_2$). For a constant force, the Lorentz equation makes the electric field, $\mathbf{E_{r_1}}$, a function of $\mathbf{B_{r_2}}$. Combining with equation 80:

$$\begin{aligned} &\mathbf{E_{r_1}} = f(\mathbf{B_{r_2}}) = 2\pi k_e q/|\mathbf{r}_2|^2 \quad \wedge \quad \mathbf{E_{r_2}} = 0 \quad \wedge \quad \mathbf{E_{r_3}} = 0 \\ &\Rightarrow \quad \nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial \mathbf{r}_1^2} & \frac{\partial}{\partial \mathbf{r}_2^2} & \frac{\partial}{\partial \mathbf{r}_3^2} \\ \mathbf{E_{r_1}} & 0 & 0 \end{vmatrix} = \left(\frac{\partial \mathbf{E}_{r_3}}{\partial \mathbf{r}_2^2} - \frac{\partial \mathbf{E}_{r_2}}{\partial \mathbf{r}_3^2}\right)\hat{\mathbf{i}} + \left(\frac{\partial \mathbf{E}_{r_1}}{\partial \mathbf{r}_3^2} - \frac{\partial \mathbf{E}_{r_3}}{\partial \mathbf{r}_1^2}\right)\hat{\mathbf{j}} + \left(\frac{\partial \mathbf{E}_{r_2}}{\partial \mathbf{r}_1^2} - \frac{\partial \mathbf{E}_{r_1}}{\partial \mathbf{r}_2^2}\right)\hat{\mathbf{k}} \\ &\qquad = 0\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + \left(0 - \frac{\partial 2\pi k_e q/|\mathbf{r}_2|^2}{\partial \mathbf{r}_2^2}\right)\hat{\mathbf{k}} \quad = \quad \frac{4\pi k_e q}{|\mathbf{r}_2|^3}\hat{\mathbf{k}} \quad = \quad \frac{4\pi k_e q}{|\mathbf{r}|^3}. \end{aligned} \tag{91}$$

Combining equations 91 and 90 yields Maxwell-Faraday's law [13]:

$$\nabla \times \mathbf{E} = \frac{4\pi k_e q}{|\mathbf{r}|^3} \quad \wedge \quad \frac{\partial \mathbf{B}}{\partial t} = -\frac{4\pi k_e q}{|\mathbf{r}|^3} \quad \Rightarrow \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{92}$$

Maxwell's wave equations

From the Gauss's electric field equations 81:

$$\mathbf{E}=\frac{2\pi k_e q}{|\mathbf{r}|^2} \quad \Rightarrow \quad \nabla^2\mathbf{E}=\frac{12\pi k_e q}{|\mathbf{r}|^4}. \tag{93}$$

$$\mathbf{E}=\frac{2\pi k_e q}{|\mathbf{r}|^2} \quad \wedge \quad r=ct \quad \Rightarrow \quad \mathbf{E}=\frac{2\pi k_e q}{(ct)^2} \quad \Rightarrow \quad \frac{\partial^2\mathbf{E}}{\partial t^2}=\frac{(12\pi k_e/c^2)q}{t^4}. \tag{94}$$

$$\frac{\partial^2\mathbf{E}}{\partial t^2}=\frac{(12\pi k_e/c^2)q}{t^4} \quad \wedge \quad r=ct \quad \Rightarrow \quad \frac{\partial^2\mathbf{E}}{\partial t^2}=\frac{12\pi k_e c^2 q}{|\mathbf{r}|^4}. \tag{95}$$

$$\nabla^2\mathbf{E}=\frac{12\pi k_e q}{|\mathbf{r}|^4} \quad \wedge \quad \frac{\partial^2\mathbf{E}}{\partial t^2}=\frac{12\pi k_e c^2 q}{|\mathbf{r}|^4} \quad \Rightarrow \quad \nabla^2\mathbf{E}=\frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2}. \tag{96}$$

$$\nabla^2\mathbf{E}=\frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2} \quad \wedge \quad \varepsilon_0\mu_0=\frac{1}{4\pi k_e}\cdot\frac{4\pi k_e}{c^2}=\frac{1}{c^2} \quad \Rightarrow \quad \nabla^2\mathbf{E}=\varepsilon_0\mu_0\frac{\partial^2\mathbf{E}}{\partial t^2}, \tag{97}$$

which is Maxwell's electric field wave equation [13]. And from equation (90):

$$\frac{\partial B}{\partial t}=-\frac{4\pi k_e q}{r^3} \quad \wedge \quad r=|\mathbf{r}| \quad \Rightarrow \quad \nabla^2\mathbf{B}=\frac{12\pi k_e q}{|\mathbf{r}|^4}. \tag{98}$$

$$\mathbf{B}=\frac{2\pi k_e q}{|\mathbf{r}|^2} \quad \wedge \quad r=ct \quad \Rightarrow \quad \mathbf{B}=\frac{2\pi k_e q}{(ct)^2} \quad \Rightarrow \quad \frac{\partial^2\mathbf{B}}{\partial t^2}=\frac{(12\pi k_e/c^2)q}{t^4}. \tag{99}$$

$$\frac{\partial^2\mathbf{B}}{\partial t^2}=\frac{(12\pi k_e/c^2)q}{t^4} \quad \wedge \quad r=ct \quad \Rightarrow \quad \frac{\partial^2\mathbf{B}}{\partial t^2}=\frac{12\pi k_e c^2 q}{|\mathbf{r}|^4}. \tag{100}$$

$$\nabla^2\mathbf{B}=-\frac{12\pi k_e q}{|\mathbf{r}|^4} \quad \wedge \quad \frac{\partial^2\mathbf{B}}{\partial t^2}=\frac{12\pi k_e c^2 q}{|\mathbf{r}|^4} \quad \Rightarrow \quad \nabla^2\mathbf{B}=\frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2}. \tag{101}$$

$$\nabla^2\mathbf{B}=\frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2} \quad \wedge \quad \varepsilon_0\mu_0=\frac{1}{4\pi k_e}\cdot\frac{4\pi k_e}{c^2}=\frac{1}{c^2} \quad \Rightarrow \quad \nabla^2\mathbf{B}=\varepsilon_0\mu_0\frac{\partial^2\mathbf{B}}{\partial t^2}, \tag{102}$$

which is Maxwell's magnetic field wave equation [13].

The classic electron radius, r_e

From the unit ratios 49:

$$r=c_m m \quad \wedge \quad r=c_q q \quad \Rightarrow \quad c_m m=c_q q \quad \Rightarrow \quad m=(c_q/c_m)q. \tag{103}$$

$$m=(c_q/c_m)q \quad \wedge \quad r=c_q q \quad \Rightarrow \quad mr=(c_q^2/c_m)q^2=\frac{r_p m_p q^2}{q_p^2}. \tag{104}$$

$$m_e r_e=\frac{r_p m_p q_e^2}{q_p^2} \quad \Rightarrow \quad r_e=r_p\frac{m_p q_e^2}{m_e q_p^2}\approx 2.81794023\cdot 10^{-15}\ m, \tag{105}$$

which is classic electron radius value [17], where m_e is the electron mass, q_e is the electron charge, and r_p, m_p, and q_p are Planck units.

The fine structure constants, α and α_G

Parsimonious explanations for the fine structure electron charge coupling constant, α, [17] and the fine structure electron gravity coupling constant, α_G, are the ratio of two subtypes of a force. Where m_e is the electron mass, q_e is the electron charge, and r_p, m_p, and q_p are Planck units:

$$\frac{F_{q_e}}{F_{q_p}} = \frac{k_e q_e^2/r^2}{k_e q_p^2/r^2} = \frac{q_e^2}{q_p^2} \approx 0.0072973525643 = \alpha. \tag{106}$$

$$\frac{F_{m_e}}{F_{m_p}} = \frac{G m_e^2/r^2}{G m_p^2/r^2} = \frac{m_e^2}{m_p^2} \approx 1.751809 \cdot 10^{-45} = \alpha_G, \tag{107}$$

Rearranging the classic electron radius equation also yields α and α_G:

$$r_e = r_p \frac{m_p q_e^2}{m_e q_p^2} \quad \Rightarrow \quad \frac{r_e m_e}{r_p m_p} = \frac{q_e^2}{q_p^2} = \alpha. \tag{108}$$

$$r_e = r_p \frac{m_p q_e^2}{m_e q_p^2} \quad \Rightarrow \quad \left(\frac{r_p q_e^2}{r_e q_p^2}\right)^2 = \frac{m_e^2}{m_p^2} = \alpha_G. \tag{109}$$

The Rydberg constant, R_∞

Starting with the ratio-derived classic electron radius (105), rearrange the quantities so that electron properties are in the numerator and the wave (Planck unit) properties are in the denominator:

$$r_e = r_p \frac{m_p q_e^2}{m_e q_p^2} \quad \Rightarrow \quad \frac{r_e}{r_p} = \frac{m_p q_e^2}{m_e q_p^2} \quad \Rightarrow \quad \frac{r_e}{r_p}\left(\frac{\alpha_G \alpha}{4\pi r_p}\right) = \frac{m_p}{m_e}\frac{q_e^2}{q_p^2}\left(\frac{\alpha_G \alpha}{4\pi r_p}\right)$$

$$\Rightarrow \quad \frac{r_e}{4\pi r_p^2}\frac{m_e^2 q_e^2}{m_p^2 q_p^2} = \frac{}{4\pi r_p}\frac{m_e q_e^4}{m_p q_p^4} \approx 1.09737315 \cdot 10^7 \ m^{-1} = R_\infty, \tag{110}$$

where m_e is the electron mass, q_e is the electron charge, and r_p, m_p, and q_p are Planck units.

$\hbar$, h, and Planck-Einstein relation

Applying both the direct proportion ratio (51), and inverse proportion ratio (49):

$$m = k_m/r \quad \wedge \quad r = ct \quad \Rightarrow \quad m(ct)^2 = (k_m/r)r^2 = k_m r. \tag{111}$$

$$m(ct)^2 = k_m r \quad \Rightarrow \quad E = mc^2 = k_m r/t^2. \tag{112}$$

$$E = mc^2 = k_m r/t^2 \ \wedge \ r/t = c \ \Rightarrow \ E = mc^2 = k_m c/t. \tag{113}$$

Solving the fine structure constant equation 108 for $r_p m_p$, where $r_p m_p = k_m$:

$$r_e m_e / r_p m_p = q_e^2 / q_p^2 \approx 0.0072973525705 = \alpha$$

$$\Rightarrow \quad k_m = r_p m_p = r_e m_e / \alpha \approx 3.5176728259 \cdot 10^{-43} \ m \cdot kg. \tag{114}$$

$$k_m c \approx 1.0545717829 \cdot 10^{-34} \ k \cdot m^2 \cdot s^{-1} \approx \hbar. \tag{115}$$

$$E = mc^2 = k_m c/t \ \wedge \ k_m c = \hbar \ \Rightarrow \ E = mc^2 = \hbar/t \ kg \cdot m^2 \cdot s^{-2} = \hbar/t \ J. \tag{116}$$

$$E = mc^2 = \hbar/t \quad \wedge \quad \omega \stackrel{\text{def}}{=} 2\pi/t \quad \wedge \quad h \stackrel{\text{def}}{=} 2\pi\hbar$$

$$\Rightarrow \quad E_\lambda \stackrel{\text{def}}{=} 2\pi E = 2\pi mc^2 = \hbar(2\pi/t) = \hbar\omega = (2\pi\hbar)/t = h/t. \tag{117}$$

Values of the unit ratios and Planck units

Values of the direct proportion ratios, c_t, c_m, and c_q:

$$c_t = c = 2.99792458 \cdot 10^8 \ m \cdot s^{-1}. \tag{118}$$

$$G = c_m c_t^2 \quad \wedge \quad G \approx 6.67430(15) \cdot 10^{-11} \ m^3 \cdot kg^{-1} \cdot s^{-2} \quad \Rightarrow$$

$$c_m = G/c_t^2 \approx 7.426162 \cdot 10^{-28} \ m \cdot kg^{-1}. \tag{119}$$

$$k_e = c_q^2 c_t^2 / c_m \quad \wedge \quad k_e \approx 8.9875517923(13) \cdot 10^9 \ N \cdot m^2 \cdot C^{-2} \quad \Rightarrow$$

$$c_q = \sqrt{k_e c_m / c_t^2} \approx 8.6175182 \cdot 10^{-18} \ m \cdot C^{-1}. \tag{120}$$

From equation 116, the values of the inverse proportion ratios, k_t, k_m, and k_q:

$$k_m = r_p m_p = r_e m_e / \alpha \approx 3.5176728259 \cdot 10^{-43} \ m \cdot kg. \tag{121}$$

$$r_p = \sqrt{r_p^2} = \sqrt{c_m k_m} \approx 1.616255 \cdot 10^{-35} \ m. \quad \Rightarrow \tag{122}$$

$$k_t = r_p^2 / c_t \approx 8.71363 \cdot 10^{-79} \ m \cdot s. \tag{123}$$

$$k_q = r_p^2 / c_q \approx 3.031361 \cdot 10^{-53} \ m \cdot C. \tag{124}$$

Values of the Planck units, r_p, t_p, m_p, and q_p::

$$r_p = \sqrt{c_t k_t} = \sqrt{c_m k_m} = \sqrt{c_q k_q} \approx 1.616255 \cdot 10^{-35} \ m. \tag{125}$$

$$t_p = r_p / c_t \approx 5.391246 \cdot 10^{-44} \ s. \tag{126}$$

$$m_p = r_p / c_m \approx 2.176434 \cdot 10^{-8} \ kg. \tag{127}$$

$$q_p = r_p / c_q = q_e / \sqrt{\alpha} \approx 1.875546038 \cdot 10^{-18} \ C. \tag{128}$$

Compton wavelength, λ

From equations 51 and 116, where the wavelength $\lambda = 2\pi r$ [13, 18]:

$$rm = k_m \quad \wedge \quad k_m c = \hbar \quad \Rightarrow \quad \lambda = 2\pi r = \frac{2\pi k_m}{m} = \frac{2\pi k_m}{m}\frac{c}{c} = \frac{2\pi\hbar}{mc} = \frac{h}{mc}. \tag{129}$$

Schrödenger's wave equation

Start with the previous unit ratio-derived "base" Planck-Einstein relation 116, where $V(r,t)$ is the potential energy in a volume. And multiply the kinetic energy component by mc/mc:

$$\forall\, V(r,t) : V(r,t) = E/2 \quad \wedge \quad E = mc^2 = \hbar/t \quad \Rightarrow \quad \hbar/t = \hbar/2t + V(r,t). \tag{130}$$

$$\hbar/t = h/2t + V(r,t) \quad \Rightarrow \quad \hbar/t = (\hbar/2t)(mc/mc) + V(r,t). \tag{131}$$

And from the distance-to-time (speed of light) ratio (49):

$$\hbar/t = \hbar mc/2mct + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar mc^2/2mcr + V(r,t). \tag{132}$$

$$\hbar/t = \hbar mc^2/2mcr + V(r,t) \;\wedge\; E = mc^2 = \hbar/t \quad \Rightarrow \quad \hbar/t = \hbar^2/2mcrt + V(r,t). \tag{133}$$

$$\hbar/t = \hbar^2/2mcrt + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar^2/2mr^2 + V(r,t). \tag{134}$$

Multiply both sides of equation 134 by a probability density function, $\Psi(r,t)$.

$$\hbar/t = \hbar^2/2mr^2 + V(r,t) \quad \Rightarrow \quad (\hbar/t)\Psi(r,t) = (\hbar^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t). \tag{135}$$

$$\begin{gathered} \forall\ \Psi(r,t) :\ \partial\Psi(r,t)/\partial t = (i\ 1/t)\Psi(r,t) \quad \wedge \quad \partial^2\Psi(r,t)/\partial r^2 = (-1/r^2)\Psi(r,t) \\ \wedge \quad (\hbar/t)\Psi(r,t) = ((\hbar)^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t) \\ \Rightarrow \quad i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t), \end{gathered} \tag{136}$$

which is the one-dimensional position-space Schrödenger's equation [3, 18].

Dirac's wave equation

Start with the previous unit ratio-derived "base" Planck-Einstein relation 116:

$$\hbar/t = mc^2 \quad \Leftrightarrow \quad \hbar/2t + \hbar/2t. = mc^2 \quad \Leftrightarrow \quad \hbar/2t = -\hbar/2t + mc^2. \tag{137}$$

$$\hbar/2t = -\hbar/2t + mc^2 \ \wedge \ qc_q/r = qc_q/ct = 1 \ \Rightarrow \ \hbar qc_q/2ct^2 = -\hbar qc_q/2rt + mc^2. \tag{138}$$

$$\hbar qc_q/2ct^2 = -\hbar qc_q/2rt + mc^2 \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar qc_q/2ct^2 = -\hbar cqc_q/2r^2 + mc^2. \tag{139}$$

$$\hbar qc_q/2ct^2 = -\hbar cqc_q/2r^2 + mc^2 \quad \Rightarrow \quad \hbar(qc_q/2ct^2)\boldsymbol{\psi} = -\hbar c(qc_q/2r^2)\boldsymbol{\psi} + mc^2\boldsymbol{\psi}. \tag{140}$$

$$\boldsymbol{\psi} = e^{iqc_q((1/2ct)-(1/2r))} \ \Rightarrow \ \partial\boldsymbol{\psi}/\partial t = (-iqc_q/2ct^2)\boldsymbol{\psi} \ \wedge \ \nabla\boldsymbol{\psi} = (iqc_q/2r^2)\boldsymbol{\psi}. \tag{141}$$

$$\begin{gathered}\partial\boldsymbol{\psi}/\partial t = (-iqc_q/2ct^2)\boldsymbol{\psi} \quad \wedge \quad \nabla\boldsymbol{\psi} = (iqc_q/2r^2)\boldsymbol{\psi} \\ \wedge \quad \hbar(qc_q/2ct^2)\boldsymbol{\psi} = -\hbar c(qc_q/2r^2)\boldsymbol{\psi} + mc^2\boldsymbol{\psi} \quad \Rightarrow \quad i\hbar\partial\boldsymbol{\psi}/\partial t = -i\hbar c\nabla\boldsymbol{\psi} + mc^2\boldsymbol{\psi}.\end{gathered} \tag{142}$$

$$\begin{gathered}i\hbar\partial\boldsymbol{\psi}/\partial t = -i\hbar c\nabla\boldsymbol{\psi} + mc^2\boldsymbol{\psi} \\ \Rightarrow \quad \exists\, \boldsymbol{\alpha}, \beta : \ i\hbar\partial\boldsymbol{\psi}/\partial t = i\hbar c\boldsymbol{\alpha} \cdot -\nabla\boldsymbol{\psi} + mc^2\beta\boldsymbol{\psi},\end{gathered} \tag{143}$$

which is the relativistic, free particle Dirac equation, where $\beta = \gamma^0$, $\boldsymbol{\alpha} = \gamma^0\gamma^k$, $k \in \{1, 2, 3\}$, and each γ a matrix [16].

Total of a type

Applying both the direct (49) and inverse proportion ratios (51) to the Euclidean distance:

$$\begin{gathered}r = \sqrt{r_1^2 + r_2^2}, \quad r_1 = c_\tau\tau, \quad r_2 = k_\tau/\tau, \quad \tau \in \{t, m, q\} \\ \Rightarrow \quad r = \sqrt{(c_\tau\tau)^2 + (k_\tau/\tau)^2} \quad \Leftrightarrow \quad \tau = \sqrt{(r/c_\tau)^2 + (k_\tau/r)^2}.\end{gathered} \tag{144}$$

Quantum extension to Schwarzschild's metric

The simplest way to demonstrate how to add quantum physics to general relativity is by extending Schwarzschild's gravitational time dilation equation and black hole metric. Apply the total of a type equation 144 to the derivation of Schwarzschild's time dilation and metric

(64):

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-(v^2/c^2)(r/r)} \quad \wedge \quad r = \sqrt{(c_m m)^2 + (k_m/m)^2} = Q_m$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-Q_m v^2/rc^2}. \tag{145}$$
$$\sqrt{1-(v^2/c^2)} = \sqrt{1-Q_m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv_{escape}^2$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m v_{escape}^2/rc^2}. \tag{146}$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m v_{escape}^2/rc^2}. \wedge v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m c^2/rc^2} = \sqrt{1-2Q_m/r}. \tag{147}$$

Combining equation 147 with equation 59 yields Schwarzschild's gravitational time dilation with a quantum mass effect:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m/r} \quad \wedge \quad t' = t\sqrt{1-(v^2/c^2)} \quad \Rightarrow \quad t' = t\sqrt{1-2Q_m/r}. \tag{148}$$

Schwarzschild defined the black hole event horizon radius, $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$. The radius with the quantum extension is $\alpha \stackrel{\text{def}}{=} 2Q_m$. At this point the exact same equations 69 through 73 yield what looks like the same Schwarzschild black hole metric.

Quantum extension to Newton's gravity force

The quantum mass effect is easier to understand in the context Newton's gravity equation than in general relativity, because the metric equations and solutions in the EFEs are much more complex. From equations 144 and 49:

$$m/\sqrt{(r/c_m)^2 + (k_m/r)^2} = 1 \quad \wedge \quad r^2/(ct)^2 = 1 \quad \Rightarrow \quad r^2/(ct)^2 = m/\sqrt{(r/c_m)^2 + (k_m/r)^2}$$
$$\Rightarrow \quad r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2}. \tag{149}$$
$$r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2} \quad \Rightarrow \quad (m/r)(r^2/t^2) = (m/r)(c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2})$$
$$\Rightarrow \quad F \stackrel{\text{def}}{=} mr/t^2 = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{150}$$
$$F = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2} \quad \wedge \quad \forall\, m \in \mathbb{R},\ \exists\, m_1, m_2 \in \mathbb{R} :\ m_1 m_2 = m^2$$
$$\Rightarrow \quad F = c^2 m_1 m_2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{151}$$

Quantum extension to Coulomb's charge force

$$q^2/((r/c_q)^2+(k_q/r)^2)=1 \;\wedge\; r^2/(ct)^2=1 \;\Rightarrow\; r^2/(ct)^2=q^2/((r/c_q)^2+(k_q/r)^2)$$
$$\Rightarrow \quad r^2/t^2=c^2q^2/((r/c_q)^2+(k_q/r)^2). \tag{152}$$

$$r^2/t^2=c^2q^2/((r/c_q)^2+(k_q/r)^2) \quad \wedge \quad m/r=1/c_m$$
$$\Rightarrow \quad F \overset{\text{def}}{=} mr/t^2=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2). \tag{153}$$

$$\forall\, q\in\mathbb{R}:\; \exists\, q_1,q_2\in\mathbb{R}:\; q_1q_2=q^2 \quad \wedge \quad F=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2)$$
$$\Rightarrow \quad F=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2). \tag{154}$$

INSIGHTS AND IMPLICATIONS

1. In this article, distance equations were derived from the principle that a distance measure, d, is an inverse (bijective) function of a volume, v, where v is the sum of subset volumes (Euclidean or non-Euclidean), $g(s_{i,1},\cdots,s_{i,n})\geq 0$. Notionally:

$$\forall\, n\in\mathbb{N},\;\; f,g\geq 0,\;\; \exists\, d:\;\; v=f(d),\;\; d=f^{-1}(v)=f^{-1}(\textstyle\sum_{i=1}^m g(s_{i,1},\cdots,s_{i,n})): \tag{155}$$

   This definition is more useful for physics than the definition of metric space.

2. The derivation of inner product, in this article, from the $n=2$ case of distance being the function of a volume that is the sum of subset volumes (16):

$$d^2=\textstyle\sum_{i=1}^m(\prod_{j=1}^{n=2}s_{i,j}) \quad \wedge \quad \exists\, a_{i,1},a_{i,2}\in\mathbb{R}:\;\; s_{i,1}=a_i,\;\; s_{i,2}=b_i$$
$$\Rightarrow \quad d^2=\sum_{i=1}^m a_ib_i \overset{\text{def}}{=} \mathbf{a}\cdot\mathbf{b}, \tag{156}$$

   is a simple explanation. In contrast, simply defining the inner product also requires inventing other arbitrary definitions, like the Kronecker's delta function, where:

$$\textstyle\sum_{i=1}^m \delta_{ij}a_ib_j=\sum_{i=1}^m a_ib_i \overset{\text{def}}{=} \mathbf{a}\cdot\mathbf{b}, \quad \delta_{ij}=\begin{cases}0 & \text{if } i\neq j,\\ 1 & \text{if } i=j.\end{cases}. \tag{157}$$

3. The left side of the distance sum inequality (.7),

$$(\textstyle\sum_{i=1}^m(a_i^n+b_i^n))^{1/n}\leq(\sum_{i=1}^m a_i^n)^{1/n}+(\sum_{i=1}^m b_i^n)^{1/n}, \tag{158}$$

differs from the left side of Minkowski's sum inequality [9].

$$(\sum_{i=1}^{m}(a_i^n + b_i^n)^{\mathbf{n}})^{1/n} \leq (\sum_{i=1}^{m} a_i^n)^{1/n} + (\sum_{i=1}^{m} b_i^n)^{1/n}. \tag{159}$$

(a) The two inequalities are only the same where $n = 1$.

(b) The distance sum inequality (.7) is a more fundamental inequality because the proof does not require the convexity and Hölder's inequality assumptions required to prove the Minkowski sum inequality [9].

(c) $\forall\ n > 1,\ v > 0$: The left side of the distance sum inequality, $v_i^n = a_i^n + b_i^n$: $\Rightarrow d = v^{1/n} = (\sum_{i=1}^{m} v_i^n)^{1/n}$, is the Minkowski distance, which makes the distance sum inequality directly related to geometry. But the left side of the Minkowski sum inequality, $d = v^{1/n} = (\sum_{i=1}^{m}((v_i^n)^{\mathbf{n}}))^{1/n} = (\sum_{i=1}^{m} v_i^{\mathbf{n^2}})^{1/n}$, is *not* a Minkowski distance.

(d) The distance sum inequality might be useful for comparing distances between concepts in machine learning.

4. *Combinatorics*: The number of ordered combinations (n-tuples) of the elements of disjoint sets was proven to imply the Euclidean volume equation (.1). And the notion of distance was derived as the inverse function of a volume. The volume and distance equations were derived without relying on the geometric primitives and relations in Euclidean geometry [10, 11], axiomatic geometry [19–23], trigonometry [24, 25], calculus [24, 26, 27], and vector analysis [2],

5. *Combinatorics*: The combinatorial AACL property (.16) implies at most 3 distance dimentsions and 3 non-distance dimension, where $\subset \mathbb{R}$ is the common property.

   (a) If there are more than 3 distance and 3 non-distance dimensions $\subseteq \mathbb{R}$, then the additional dimensions are either not commutative or not linear sequenceable.

   (b) There cannot be an AACL of more than 3 quantum states of the same type, for example, 3 charge colors and 3 anti-charge colors.

6. *Combinatorics*: The combinatorial AACL property (.16) implies that for each constant unit interval length, r_p, of a 3-dimensional distance, there are only 3 possible correspondences to non-distance constant unit interval lengths: 3 direct constant proportion unit ratios (49): $r = (r_p/t_p)t = (r_p/m_p)m = (r_p/q_p)q$ and 3 inverse constant

proportion ratios (51): $r = (r_p t_p)/t = (r_p m_p)/m = (r_p q_p)/q$, where r_p, t_p, m_p, and q_p are the Planck units (125).

7. A discrete (point) valued state has measure 0 (zero-length interval). The ratio of a time, distance, mass, or charge interval length to zero is undefined, which is the reason discrete, state values exist independent of distance and time. in other words, discrete state values, including entangled state values, change instantly over any distance.

8. In this article, the following constants are derived directly from the ratios and Planck units: gravity, $G = c_m c_t^2 = (r_p/m_p)(r_p/t_p)^2 = r_p^3/m_p t_p^2$ (54), reduced Planck $\hbar = k_m c_t = (r_p m_p)(r_p/t_p) = r_p^2 m_p/t_p$ (116), charge, $k_e = c_q^2 c_t^2/c_m$ (77), vacuum permittivity $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$ (81), and vacuum permeability $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m$ $= 4\pi r_p m_p/q_p^2 = 4\pi k_m/q_p^2$ (85).

   And the quantum extensions to: Schwarzschild's gravitational time dilation (147), Newton's gravity force (151), and Coulomb's charge force (154), show that the constants G, $\kappa = 8\pi G/c^4$, k_e, and ε_0 do not exist, are decomposed, where the quantum effects are measurable.

   This provides the insight that the constants: G, κ, k_e, ε_0, and μ_0 are *not* fundamental constants because 1) they were all shown to be composed of the Planck units and 2) G, κ, k_e, and ε_0 do not exist, are decomposed, where the quantum effects are measurable.

9. c_t, c_m, and c_q are the *maximum* proportionality constants: $r = \sqrt{r_1^2 + r_2^2 + r_3^2} \;\Rightarrow\; r_1,\ r_2,\ r_3 \leq r \;\Rightarrow\; \forall\, \tau \in \{t, m, q\} : r_1/\tau,\ r_2/\tau,\ r_3/\tau \leq r/\tau = r_p/\tau_p = c_\tau$. For example, the speeds of gravity and light waves are limited to the proportionality constant, c_t.

10. c_t (the speed of light) is a component of the constants: $G = c_m c_t^2$, $k_e = c_q^2 c_t^2/c_m$, $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$, and $\hbar = k_m c_t$. The constants, derived in this article, that do not contain c_t are: vacuum permeability: $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m$, the fine structures, $\alpha = q_e^2/q_p^2$, and $\alpha_G = m_e^2/m_p^2$, and Rydberg, $R_\infty = m_e q_e^4/4\pi r_p m_p q_p^4$.

11. An empirical law and related constant *describes* the relations between the variables in an equation, where the units of the constant are added ad hoc to make the units on both sides of an equation balance. In contrast, where an equation and related

constant are derived from the ratios, the ratios are the *reason/explanation* for the relations between the variables and the *reason/explanation* for the constant's units.

The constant interval lengths, r_p, t_p, m_p, and q_p, have units (dimensions). Therefore, the ratios and all equations and constants derived from the ratios have dimensions, which eliminates the need for dimensional analysis. For example, $G = c_m c_t^2 = r_p^3/m_p t_p^2$, which conforms to the SI units: $m^3 \cdot kg^{-1} \cdot s^{-2}$ [17].

12. The ratio-based derivation of Lorentz's law (87), here, differs from other relativistic derivations, in that equations for the magnetic field, $\mathbf{B}$, and magnetic permeability, μ_0, are derived in the process. The ratio-derived equation, $\mathbf{B} = (2\pi k_e/c^2)vq/|\mathbf{r}|^2$, allows a simple derivation of the Faraday's law (92) and the magnetic component of Maxwell's electromagnetic equations (102).

13. Other derivations of Maxwell's electric and magnetic field wave equations require: Gauss's electric field law in a vacuum: $\nabla \cdot \mathbf{E} = -\rho/\varepsilon_0$, Gauss's magnetic field law in a vacuum: $\nabla \cdot \mathbf{B} = 0$, Faraday's Law: $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$, and Ampere's Law: $\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \partial \mathbf{E}/\partial t$ [13]. In contrast, the derivations (97) (102), in this article, did not require any of those laws. Just the speed of light ratio, $r = ct$ and ratio-derived definitions of $\mathbf{E}$ and $\mathbf{B}$ are used.

14. As shown in the subsection 64, deriving the Schwarzschild's gravitational time dilation and black hole metric from ratios is a simple way, that in *some* cases, can be used to derive an analytic solution that does not require the complexity of manipulating Christoffel symbols, using determinants, etc. [14, 15]

15. Einstein's relativity equations: 1) assume the Lorentz transformations, 2) assume the laws of physics are same at each local Euclidean-like coordinate point, 3) assume the notion of light, and 4) assume that the speed of light is the same at each coordinate point [1, 2].

    The derivations, in this article, were made without those assumptions (does not even require the notion of light). The unit ratios are *constant* ratios of a *constant* Euclidean distance interval length to *constant* interval lengths of time, mass, and charge. Therefore, the *constant* unit ratios and all equations (laws) and constants derived from those ratios and Euclidean distance are the same at each local Euclidean-like coordinate point.

16. Some common assumptions about the reduced and full Planck constants are incorrect!

    (a) The reduced Planck constant, $\hbar$, was shown to be composed of the ratio (proportionality) constants, k_m and c_t, where: $k_m c_t = \hbar$. Therefore, the common assumption that the Planck constant is an "atomic" constant is incorrect!

    (b) The derivation of the Planck-Einstein equation (116) shows that: $E = mc^2 = \hbar/t \Rightarrow E_\lambda = 2\pi E = 2\pi mc^2 = \hbar(2\pi/t) = \hbar\omega = (2\pi\hbar)/t = h/t$. Therefore, the common assumption that $E = mc^2 = \hbar\omega = h/t$ is incorrect!

    Further, the derivations, in this article, of Schrödenger's wave equation (136) [3] and Dirac's wave equation (143) [16] from the ratio-derived Planck-Einstein relation, $E = mc^2 = \hbar/t$ (116), are shorter, simpler, and require fewer assumptions.

    Therefore, it would be more logical for CODATA to define the reduced Planck constant as $\hbar = k_m c_t$ and define the full Planck constant in terms of the reduced Planck constant, $h = 2\pi\hbar$.

17. In the ratio-based derivation of Schrödenger's wave equation (136) [3], more work is needed to determine what probability distaribution function, $\psi(r,t)$, satisfies the partial derivative constraints, $\partial\Psi(r,t)/\partial t = (i\ 1/t)\Psi(r,t)$ and $\partial^2\Psi(r,t)/\partial r^2 = (-1/r^2)\Psi(r,t)$.

18. The complicated and obtuse CODATA definition, $\alpha = q_e^2/4\pi\varepsilon_0\hbar c$, [17], derived from dimensional analysis, reduces to the simpler, ratio-derived equation, in this article:

$$\varepsilon_0 = 1/4\pi k_e = 1/(4\pi(c_q^2 c_t^2/c_m) \quad \wedge \quad \hbar = k_m c_t \quad \wedge \quad h = 2\pi\hbar$$
$$\Rightarrow \quad \varepsilon_0 hc = 2\pi k_m c_t^2/(4\pi(c_q^2/c_m)c_t^2) = k_m/(2(c_q^2/c_m))$$
$$= m_p r_p/(2((r_p/q_p)^2/(r_p/m_p))) = q_p^2/2. \quad (160)$$

$$\alpha = q_e^2/2\varepsilon_0 hc \quad \wedge \quad \varepsilon_0 hc = q_p^2/2 \quad \Rightarrow \quad \alpha = q_e^2/2(q_p^2/2) = q_e^2/q_p^2. \quad (161)$$

19. The complicated and obtuse fine structure electron gravity coupling constant [17], derived from dimensional analysis, reduces to the simpler, ratio-derived equation:

$$\alpha_G = Gm_e^2/\hbar c = (c_m c_t^2)m_e^2/(k_m c_t)c_t = (r_p/m_p)m_e^2/m_p r_p = m_e^2/m_p^2. \quad (162)$$

20. The complicated and obtuse NIST/CODATA definition of the vacuum permeability constant, $\mu_0 = 2h\alpha/cq_e^2$ [17], (derived from dimensional analysis) reduces to a simpler ratio-derived definition. Using the ratios to reduce the old definition of μ_0 derived in this article:

$$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2. \tag{163}$$

Using the ratios to reduce the CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$ [17]:

$$h = 2\pi k_m c_t, \;\; \alpha = q_e^q/q_p^2 \;\; \Rightarrow \;\; \mu_0 = 2h\alpha/cq_e^2 = 4\pi k_m c_t(q_e^2/q_p^2)/c_t q_e^2 = 4\pi k_m/q_p^2. \tag{164}$$

21. The obtuse NIST/CODATA definition of the Rydberg constant [17], derived from dimensional analysis, reduces to the more informative ratio-derived equation (110):

$$\begin{aligned} R_\infty = \frac{m_e q_e^4}{8\varepsilon_0^2 h^3 c} &= \frac{m_e q_e^4}{8(1/4\pi k_e)^2(2\pi\hbar)^3 c_t} = \frac{m_e q_e^4(4\pi k_e)^2}{8(2\pi\hbar)^3 c_t} = \frac{m_e q_e^4(c_q^2 c_t^2/c_m)^2}{4\pi(k_m c_t)^3 c_t} \\ &= \frac{m_e q_e^4(c_q^2/c_m)^2}{4\pi k_m^3} = \frac{m_e q_e^4 c_q^4}{4\pi k_m^3 c_m^2} = \frac{m_e q_e^4(r_p/q_p)^4}{4\pi(r_p m_p)^3(r_p/m_p)^2} = \frac{}{4\pi r_p}\frac{m_e q_e^4}{m_p q_p^4}. \end{aligned} \tag{165}$$

22. The quantum extensions to: Schwarzschild's black hole metric (73), Newton's gravity force (151), and Coulomb's charge force (154) make quantifiable predictions:

    (a) For gravity, $\lim_{r\to 0} F = c^2 m_1 m_2/k_m$, and for charge, $\lim_{r\to 0} F = 0$. Finite maximum gravity and charge forces eliminates the problem of forces going to infinity as $r \to 0$.

    (b) The quantum-extended relativity and classic equations reduce to the non-extended relativity and classic equations and constants, where the distance between 2 masses and between 2 charges is sufficiently large or the masses and charges sufficiently large that the quantum effects are not measurable. *Note* that G, k_e, ε_0, μ_0, and κ (Einstein's constant, which contains G) do not exist, are decomposed, where the quantum effects becomes measurable.

    (c) $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$ implies that as distance $\to 0$, spacetime curvature peaks at the Planck unit length (distance), r_p (125). The ratio of quantum units, m_p/r_p^3, might indicate a maximum mass density (where only approximately 10^{19} protons and neutrons can be compressed into a 1 cubic Planck length volume). A finite force (spacetime curvature) and finite mass density might imply that black holes have sizes > 0 (are not singularities), which might impact whether black holes evaporate and whether a big bang would have originated from a singularity.

DATA AVAILABILITY STATEMENT

The supplemental files are computer source code files (ASCII text) expressing formal math proofs used to validate the proofs in this article. The source code files are compiled by a logic engine called Rocq (formerly known as Coq) [8]. If the files compile without error, then the consensus of many in the mathematics community is that the proofs are highly likely to be correct.

The supplemental file, "euclidrelations.v," contains formal proofs about Euclidean volume, inner product, Minkowski distance, distance inequalities, and metric space. The supplemental file, "threed.v," contains a formal proof of the properties that limit set to at most 3 elements.

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